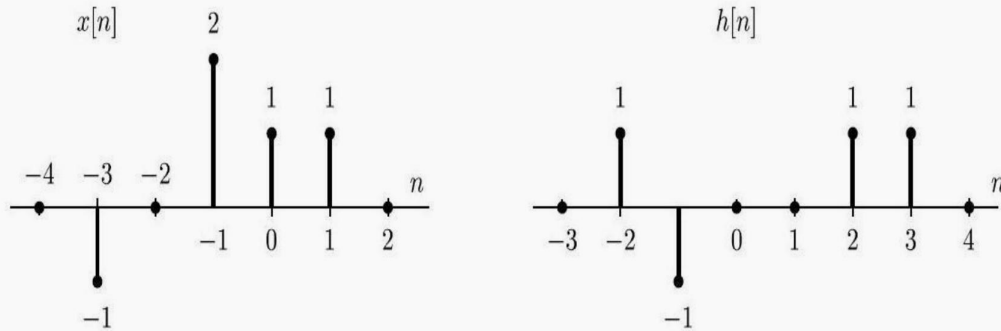


### Problem 1

Compute the convolution  $y[n] = x[n] * h[n]$  of each of the two following pairs of signals:

(a).  $x[n]$  and  $h[n]$  are depicted below



(b).  $x[n] = u[n+4] - u[n-1]$ ,  $h[n] = 2^n u[2-n]$ .

Problem 1 (a)  $y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$

$\Rightarrow y[n] = \{\dots, 0, \overset{-5}{-1}, 1, 2, -2, -1, 1, 3, 2, 1, 0, \dots\}$

(b)  $x[n] = u[n+4] - u[n-1]$   $\forall n \in [-4, 0]$  时  $x[n] = 1$

$\Rightarrow y[n] = \sum_{k=-4}^0 x[k] h[n-k] = \sum_{k=-4}^0 h[n-k]$

$n-k \leq 2$  时,  $y[n] = h[n-k] = 2^{n-k}$  此时  $k \geq n-2$

①  $n \leq -2$  时,  $k \geq n-2$  恒成立.  $y[n] = \sum_{k=-4}^0 2^{n-k} = 2^n \sum_{k=-4}^0 2^{-k} = 31 \cdot 2^n$ .

②  $n \in [-2, 2]$  时,  $y[n] = \sum_{k=n-2}^0 2^{n-k} = 2^n \sum_{k=n-2}^0 2^{-k} = 2^n (2^{3-n} - 1) = 8 - 2^n$ .

③  $n > 2$  时,  $y[n] = 0$ .

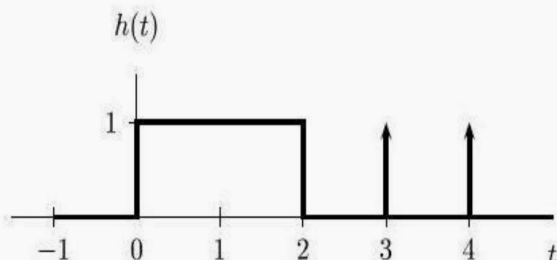
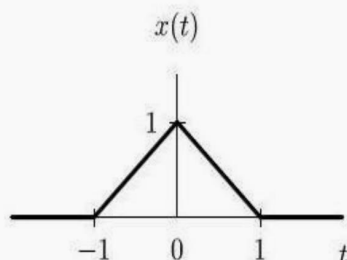
$\Rightarrow y[n] = \begin{cases} 31 \cdot 2^n, & n \leq -2 \\ 8 - 2^n, & n \in [-2, 2] \\ 0, & n > 2 \end{cases}$

## Problem 2

Compute the convolution  $y(t) = x(t) * h(t)$  for each of the following pairs of signals:

(a).  $x(t) = e^{-t}u(t+1)$ ,  $h(t) = e^{2t}u(-t)$

(b).  $x(t)$  and  $h(t)$  are depicted below:



Problem 2 (a)  $x(t) = e^{-t}u(t+1)$ ,  $h(t) = e^{2t}u(-t)$

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau = \int_{-\infty}^{\infty} e^{-\tau} u(\tau+1) e^{2(t-\tau)} u(-(t-\tau)) d\tau = e^{2t} \int_{-\infty}^{\infty} e^{-3\tau} u(\tau+1) u(\tau-t) d\tau$$

$\tau > \max\{-1, t\}$  时 为 零.

①  $t \leq -1$  时,  $y(t) = e^{2t} \int_{-1}^{\infty} e^{-3\tau} d\tau = \frac{1}{3} e^{2t+3}$

②  $t > -1$  时,  $y(t) = e^{2t} \int_t^{\infty} e^{-3\tau} d\tau = \frac{1}{3} e^{-t}$

$$\Rightarrow y(t) = \begin{cases} \frac{1}{3} e^{2t+3}, & t \leq -1 \\ \frac{1}{3} e^{-t}, & t > -1. \end{cases}$$

(b) 令  $p(t)$ :  $p(t) = \text{rect}(t)$   $h(t) = p(t) + \delta(t-3) + \delta(t-4)$

$$y(t) = x(t) * p(t) + x(t) * \delta(t-3) + x(t) * \delta(t-4)$$

$$x(t) * p(t) = \int_{-\infty}^{\infty} x(\tau) p(t-\tau) d\tau = \int_{t-2}^t x(\tau) d\tau$$

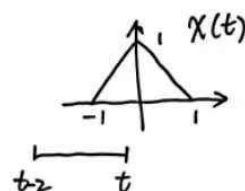
①  $t \in (-1, 0]$  时,  $\int_{t-2}^t x(\tau) d\tau = \frac{1}{2} (t+1)^2$

②  $t \in (0, 1]$  时,  $\int_{t-2}^t x(\tau) d\tau = 1 - \frac{1}{2} (1-t)^2$

③  $t \in (1, 2]$  时,  $\int_{t-2}^t x(\tau) d\tau = 1 - \frac{1}{2} (t-1)^2$

④  $t \in (2, 3]$  时,  $\int_{t-2}^t x(\tau) d\tau = 0$

$$\Rightarrow y(t) = \begin{cases} \frac{1}{2} (t+1)^2 + x(t-3) + x(t-4), & t \in (-1, 0] \\ 1 - \frac{1}{2} (1-t)^2 + x(t-3) + x(t-4), & t \in (0, 1] \\ 1 - \frac{1}{2} (t-1)^2 + x(t-3) + x(t-4), & t \in (1, 2] \\ x(t-3) + x(t-4), & t \in \text{其他} \end{cases}$$



### Problem 3

The following are impulse responses of either discrete-time or continuous-time LTI systems. Determine whether each system is causal and/or stable. Justify your answer:

(a).  $h[n] = 2^n u[3 - n]$

(b).  $h(t) = u(1 - t) - \frac{1}{2}e^{-t}u(t)$

(c).  $h[n] = [1 - (0.99)^n]u[n]$

(d).  $h(t) = e^{15t} [u(t - 1) - u(t - 100)]$

Problem 3 (a)  $h[n] = 2^n u[3 - n]$

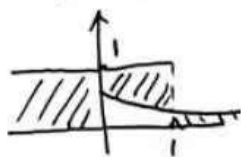


因果, 稳定.

因果  
稳定:  $t < 0$  时  $h(t) / h[n] = 0$   
稳定:  $\int_{-\infty}^{\infty} h(t) dt < \infty$

(b)  $h(t) = u(1 - t) - \frac{1}{2}e^{-t}u(t)$

在  $t < 0$  时  $h(t) \neq 0$ , 非因果



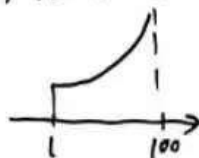
在此处不收敛, 不稳定.

(c)  $h[n] = [1 - 0.99^n]u[n]$



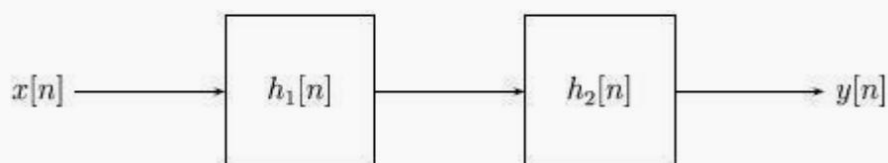
因果; 不稳定.

(d)  $h(t) = e^{15t} [u(t - 1) - u(t - 100)]$



因果; 稳定.

Consider the cascade of LTI systems with unit sample responses  $h_1[n]$  and  $h_2[n]$  depicted below:

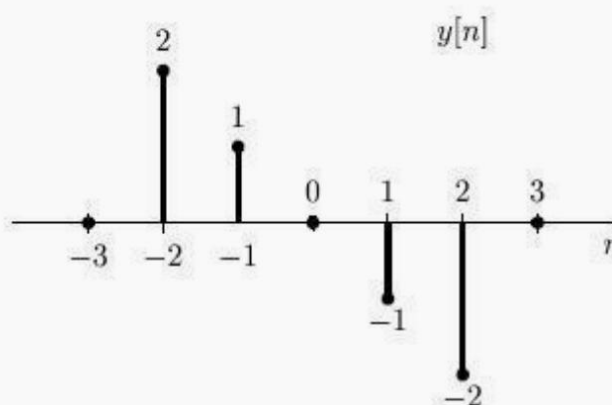


- $h_2[n] = \delta[n] - \delta[n - 1]$

- If the input is

$$x[n] = u[n] - u[n - 2]$$

then the output is as depicted below



Find  $h_1[n]$ .

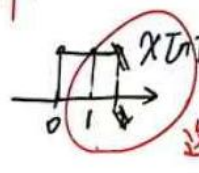
$$x[n] \xrightarrow{h_1[n]} y_1[n] \xrightarrow{h_2[n]} y[n]$$

→ 交换、分配律

$$y[n] = x[n] * h_1[n] * h_2[n] = (x[n] * h_2[n]) * h_1[n]$$

$$x[n] * h_2[n] = x[n] - x[n-1] = \begin{matrix} & 1 & \\ & | & \\ & \bullet & \\ \text{---} & 0 & \text{---} \end{matrix} \begin{matrix} 1 \\ 2 \end{matrix}$$

$$= \delta[n] - \delta[n-1]$$



$$x[n] = u[n] - u[n-2]$$

边下注意

$$y[n] = 2\delta[n+2] + \delta[n+1] - \delta[n-1] - \delta[n-2]$$

都写成  $\Gamma$  形式, 根据平移性质.

$$(\delta T_n - \delta T_{n-2}) (2\delta T_{n+2} + \delta T_{n+1} + 2\delta T_n) = 2\delta T_{n+2} - 2\delta T_{n+1} + \delta T_n - \delta T_{n-1} + 2\delta T_n - 2\delta T_{n-2} = 4T_n.$$

没有(端点): 凑时从高位凑, 系数相同.

$$+2\delta[n] - 2\delta[n-1] = 4\delta[n].$$

$$\Rightarrow h_1[n] = 2\delta[n+2] + \delta[n+1] + 2\delta[n] =$$


### Problem 5

6.1. Consider a continuous-time LTI system with frequency response  $H(j\omega) = |H(j\omega)|e^{j\angle H(j\omega)}$  and real impulse response  $h(t)$ . Suppose that we apply an input  $x(t) = \cos(\omega_0 t + \phi_0)$  to this system. The resulting output can be shown to be of the form

$$y(t) = Ax(t - t_0),$$

where  $A$  is a nonnegative real number representing an *amplitude-scaling* factor and  $t_0$  is a time delay.

- (a) Express  $A$  in terms of  $|H(j\omega_0)|$ .
- (b) Express  $t_0$  in terms of  $\angle H(j\omega_0)$ .

Problem 5 (a)  $A = |H(j\omega_0)|$

(b)  $\angle H(j\omega_0) = -\omega_0 t_0 \Rightarrow t_0 = -\frac{\angle H(j\omega_0)}{\omega_0}$

### Problem 6

6.3. Consider the following frequency response for a causal and stable LTI system:

$$H(j\omega) = \frac{1 - j\omega}{1 + j\omega}.$$

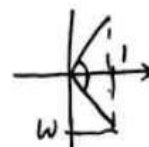
- (a) Show that  $|H(j\omega)| = A$ , and determine the value of  $A$ .
- (b) Determine which of the following statements is true about  $\tau(\omega)$ , the group delay of the system. (Note:  $\tau(\omega) = -d(\angle H(j\omega))/d\omega$ , where  $\angle H(j\omega)$  is expressed in a form that does not contain any discontinuities.)
  - 1.  $\tau(\omega) = 0$  for  $\omega > 0$
  - 2.  $\tau(\omega) > 0$  for  $\omega > 0$
  - 3.  $\tau(\omega) < 0$  for  $\omega > 0$

(a)  $|H(j\omega)| = \frac{|1 - j\omega|}{|1 + j\omega|} = \frac{\sqrt{1 + \omega^2}}{\sqrt{1 + \omega^2}} = 1$  也可以  $|\frac{1 - j\omega}{1 + j\omega}| = |A + jB|$  算. 麻烦.

(b)  $\tau(\omega) = -\frac{d\angle H(j\omega)}{d\omega}.$

$\angle H(j\omega): \angle H(j\omega) = \angle(1 - j\omega) - \angle(1 + j\omega)$

$= -2\arctan \omega.$



$\tau(\omega) = 2 \frac{d\arctan \omega}{d\omega} = \frac{2}{1 + \omega^2} > 0. \Rightarrow \tau(\omega) > 0 \text{ for } \omega > 0$

# Problem 7

6.12. A continuous-time LTI system  $S$  with frequency response  $H(j\omega)$  is constructed by cascading two continuous-time LTI systems with frequency responses  $H_1(j\omega)$  and  $H_2(j\omega)$ , respectively. Figures P6.12(a) and P6.12(b) show the straight-line approximations of the Bode magnitude plots of  $H_1(j\omega)$  and  $H(j\omega)$ , respectively. Specify  $H_2(j\omega)$ .

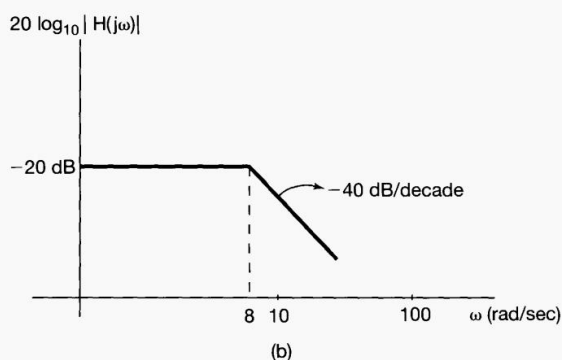
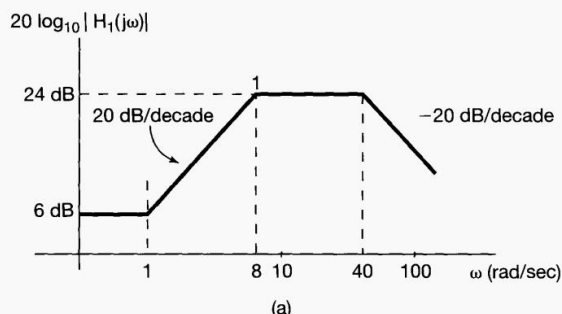


Figure P6.12

blem 7

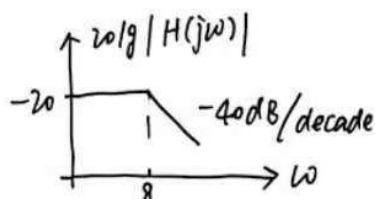
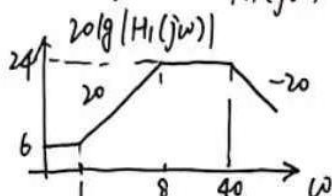
Bode plot.

$\uparrow$  dB =  $20 \log_{10} |H(j\omega)|$   
 $\rightarrow$  对数

$H(j\omega) = K \frac{(1+j\frac{\omega}{z_1})}{(1+j\frac{\omega}{p_1})(1+j\frac{\omega}{p_2})}$   
 $\downarrow$  零极点带来斜率增加 20 dB/1 格  
 $\downarrow$  极点: 斜率减小  
 $\omega=0$  时,  $20 \log K =$  纵坐标.

$$H(j\omega) = H_1(j\omega) \cdot H_2(j\omega)$$

$$\Rightarrow H_2(j\omega) = \frac{H(j\omega)}{H_1(j\omega)}$$



时域  $h(t)$ : 卷积  
 频域  $H(j\omega)$ : 乘积  
 $\downarrow$  傅立叶变换

$$20 \log K = 6 \Rightarrow K = 2$$

$$H_1(j\omega) = 2 \frac{1+j\omega}{(1+j\frac{\omega}{8})(1+j\frac{\omega}{40})}$$

$$20 \log K = -20 \Rightarrow K = 0.1$$

$$H(j\omega) = 0.1 \frac{1}{(1+j\frac{\omega}{8})^2}$$

$$H_2(j\omega) = \frac{0.1 \frac{1}{(1+j\frac{\omega}{8})^2}}{2 \frac{1+j\omega}{(1+j\frac{\omega}{8})(1+j\frac{\omega}{40})}} = 0.05 \frac{1+j\frac{\omega}{40}}{(1+j\omega)(1+j\frac{\omega}{8})}$$

# Problem 8

6.28. (a) Sketch the Bode plots for the following frequency responses:

(i)  $1 + (j\omega/10)$

(ii)  $1 - (j\omega/10)$

(iii)  $\frac{16}{(j\omega+2)^4}$

(iv)  $\frac{1-(j\omega/10)}{1+j\omega}$

(v)  $\frac{(j\omega/10)-1}{1+j\omega}$

(vi)  $\frac{1+(j\omega/10)}{1+j\omega}$

(vii)  $\frac{1-(j\omega/10)}{(j\omega)^2+(j\omega)+1}$

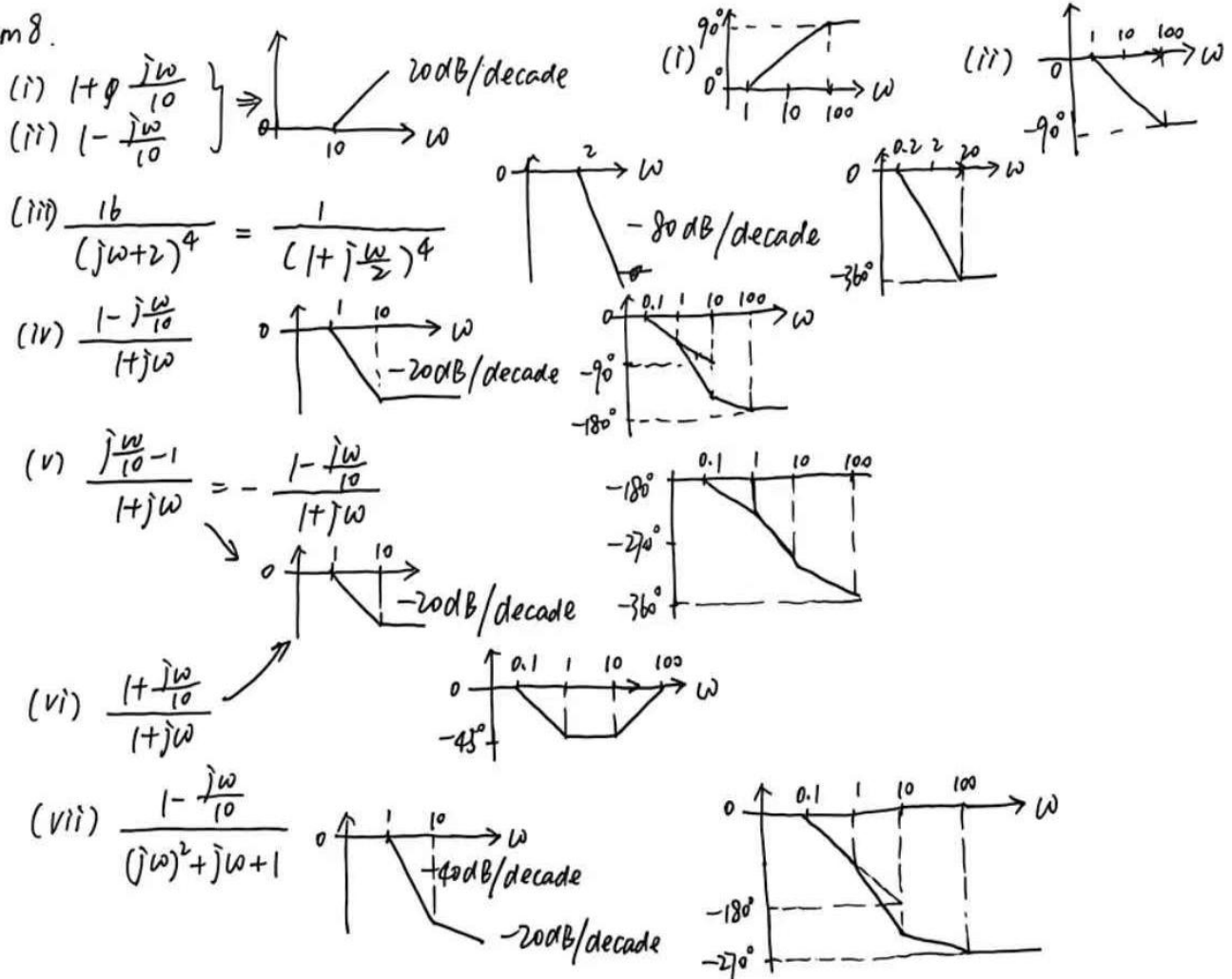
(viii)  $\frac{10+5j\omega+10(j\omega)^2}{1+(j\omega/10)}$

(ix)  $1 + j\omega + (j\omega)^2$

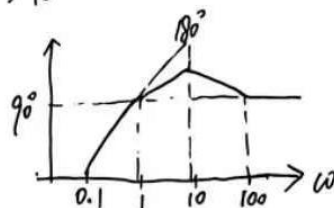
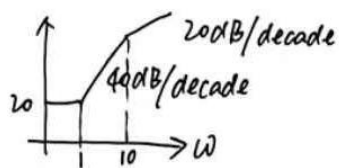
(x)  $1 - j\omega + (j\omega)^2$

(xi)  $\frac{(j\omega+10)(10j\omega+1)}{[(j\omega/100+1)][((j\omega)^2+j\omega+1)]}$

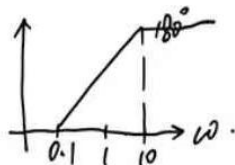
Problem 8.



$$(vii) \frac{(0.5j\omega + 10)(j\omega)^2}{1 + j\frac{\omega}{10}} = 10 \cdot \frac{1 + \frac{1}{2}j\omega + (j\omega)^2}{1 + j\frac{\omega}{10}}$$

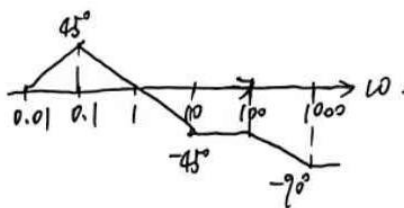
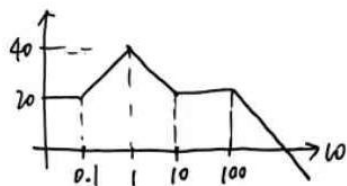


$$(ix) 1 + j\omega + (j\omega)^2$$



$$(x) 1 - j\omega + (j\omega)^2 \text{ 同上.}$$

$$(xi) \frac{(j\omega + 10)(10j\omega + 1)}{(j\frac{\omega}{100} + 1)(j\omega)^2 + j\omega + 1} = 10 \cdot \frac{(1 + j\frac{\omega}{10})(1 + 10j\omega)}{(1 + j\frac{\omega}{100})(j\omega)^2 + j\omega + 1}$$



Problem 9

6.30. Let  $h(t)$  have the Bode plot depicted in Figure P6.30. The dashed lines in the figure represent straight-line approximations. Sketch the Bode plots for  $10h(10t)$ .

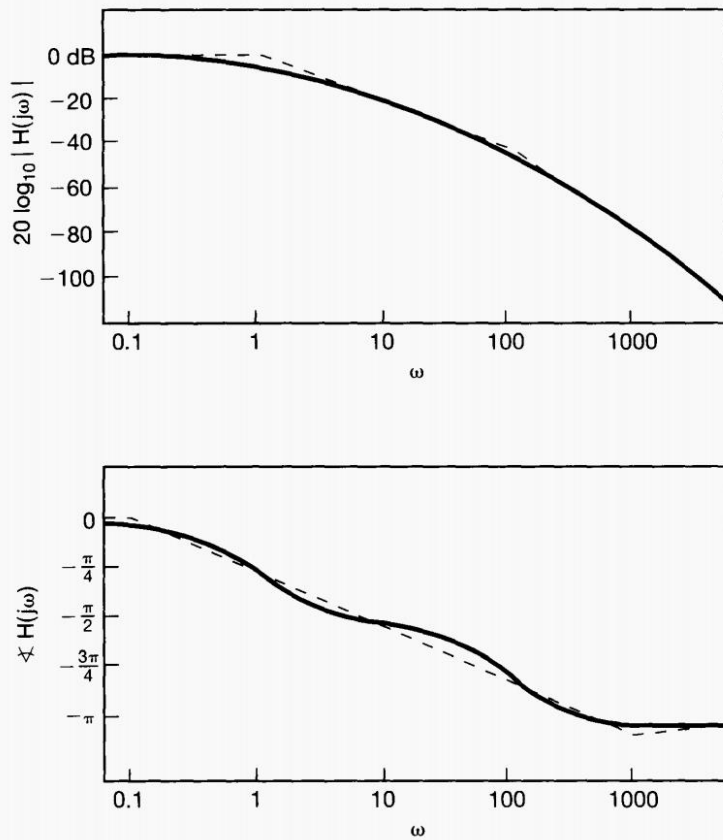


Figure P6.30

$$h(t) \xrightarrow{F} H(j\omega)$$

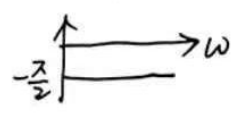
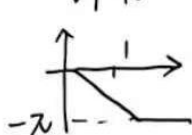
$$10h(10t) \xrightarrow{F} H(j\frac{\omega}{10})$$

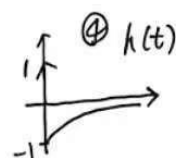
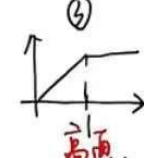
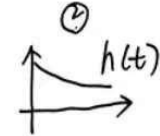
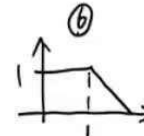
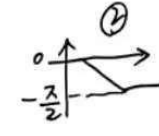
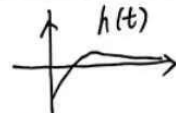
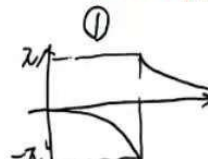
幅度图. 相位图均向右平移 1 decade.

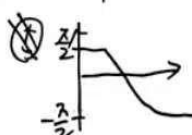
# Problem 10

The following plots show pole-zero diagrams, impulse responses, Bode magnitude plots, and Bode angle plots for six causal CT LTI systems. Determine which corresponds to which.

Problem 10 零极点图  $\rightarrow H(s) \rightarrow$   $\begin{cases} \text{时域: 冲激响应 } h(t) \\ \text{频域: Bode.} \end{cases}$

| Pole-zero diagram                                                                                                                                                 | Impulse response                                                                        | Bode Magnitude                                                                                                       | Angle                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| ① $p=0 \Rightarrow H(s)=\frac{1}{s}$                                                                                                                              | $h(t)=u(t)$<br>③                                                                        | $ H(j\omega) =\frac{1}{\omega}$<br>$20\lg\frac{1}{\omega}=-20\lg\omega$<br>⑤                                         | ④<br>                                  |
| ② $p=a\pm jb \Rightarrow H(s)=\frac{A}{s^2+cs+d}$<br>$h(0^+)=\lim_{s\rightarrow 0}\frac{As}{s^2+cs+d}=0$<br>= 振荡<br>推导得 $h(t)=Ke^{at}\sin(\omega t)u(t)$<br>衰减正弦波 | ①<br> | ②<br>$-40\text{dB/decade 下降}$<br> | ③<br>$\pi \rightarrow -180^\circ$<br> |

| Pole-zero diagram                                                                                                                                                                          | Impulse response                                                                         | $H=j\omega$             | Bode Magnitude                                                                            | Angle                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ③ $H(s)=\frac{s}{s+1}=1-\frac{1}{s+1}$<br>$\Rightarrow h(t)=\delta(t)-e^{-t}u(t)$                                                                                                          | ④<br> | $H=j\omega$             | ⑤<br> | $\angle = \angle(j\omega) - \angle(1+j\omega)$<br>$= 90^\circ - \arctan\omega$<br>⑥<br> |
| ④ $H(s)=\frac{1}{s+1}$<br>$\Rightarrow h(t)=e^{-t}u(t)$                                                                                                                                    | ②<br> | $H=\frac{1}{1+j\omega}$ | ⑥<br> | ②<br>                                                                                   |
| ⑤ $H(s)=\frac{As^2+Bs+C}{Ds^2+Es+F}$<br>$\Rightarrow H(s)=\frac{s^2-2as+(a^2+b^2)}{s^2+2as+(a^2+b^2)}=1-\frac{4as}{s^2+2as+(a^2+b^2)}$<br>镜像 $\Rightarrow h(t)=\delta(t)+\dots$<br>再相初值/终值 | ⑥<br> | $ H(j\omega) =1$ 全通     | ①<br> | ①<br>高位移 $= (-\pi) + (-\pi)$<br>极点 右半平面零点<br>                                           |

|                                                |                                                                                          |                                                                                           |                                                                                                                            |
|------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| ⑥ $H(s)=\frac{s}{As^2+Bs+C}$<br>初值: $h(0^+)=1$ | ⑤<br> | ④<br> | ⑤<br>$90^\circ$ 起, $-90^\circ$ 止.<br> |
|------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|